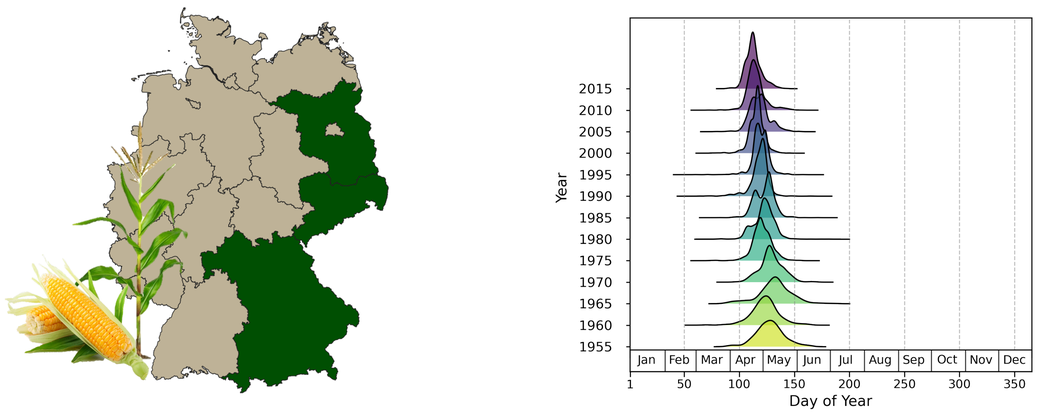
INTERNATIONAL STATISTICAL ECOLOGY CONFERENCE 2027

MODELING SOWING TIMING UNDER CLIMATE VARIABILITY USING RANDOM WALKS ON GRAPHS

Isaac Vázquez Mendoza[1], Magali Arellano Vázquez[1].
[1]INFOTEC, Centro de Investigación e Innovación en Tecnologías de la Información y Comunicación.

Introduction

The progression and duration of crop developmental stages are largely driven by environmental conditions [9]. Climate change is expected to alter the timing and duration of these stages [13, 12]. According to Baum *et al.* (2019) [2], sowing dates may account for up to 70% of maize yield variability. Depending on geographical location, maize sowing dates may remain unchanged or shift earlier or later in the year [10]. Climate change projections suggest that northern European regions, including Germany, may become increasingly suitable for maize cultivation [11]. In Germany, maize sowing dates have shifted towards earlier dates, as illustrated in Figure 1b. This shift has been associated with climate change and has intensified since the late 1980s [3]. Traditionally, phenological event timing is predicted using Growing Degree Days (GDD), an approach based primarily on air temperature [1]. In this context, we developed a probabilistic model based on four variables to predict maize sowing windows for each of the three German federal states considered in this study: Bavaria, Brandenburg, and Saxony, see Figure 1a. A region encompassing approximately 55% of the country's area devoted to major crops [4].



(a) Area of study. From top to bottom: Brandenburg, Saxony, and Bavaria.

(b) Distributions of maize sowing dates in Germany.

Figure 1: Geographical context and historical trends in maize sowing dates in Germany.

Methodology

Four raster datasets obtained from the Deutscher Wetterdienst (DWD) [5, 6, 7, 8] covering the period 1996–2025, were used as input data for the probabilistic model. These datasets provide daily measurements of mean air temperature, mean relative humidity, mean soil temperature at a depth of 5 cm, and mean soil moisture in the 0–10 cm soil layer, respectively, at a spatial resolution of 1×1 km. After a literature review, variable-specific base thresholds for each variable were given, see Table 1. For each day k, $k \in \{1, \ldots, 365\}$, a point cloud in $\mathbb{R}^4$ is constructed by gathering the data corresponding to the k-th day of each year within the period under consideration. Thus, the goodness-of-sowing of day k can be assessed by computing the distance between the centroid of this point cloud, $C_{H,k}$, and the centroid of the region defined as the Cartesian product of the intervals reported in Table 1, C^*. This region represents the suggested environmental conditions for maize sowing.

Agroclimatic variable	Interval used in this study
Mean air temperature	11.10 ± 2.51 °C
Mean relative humidity	65–85 %
Mean soil temperature at a depth of 5 cm	8–10 °C
Mean soil moisture in the 0–10 cm soil layer	80–90 %

Table 1: Base threshold per agroclimatic variable used.

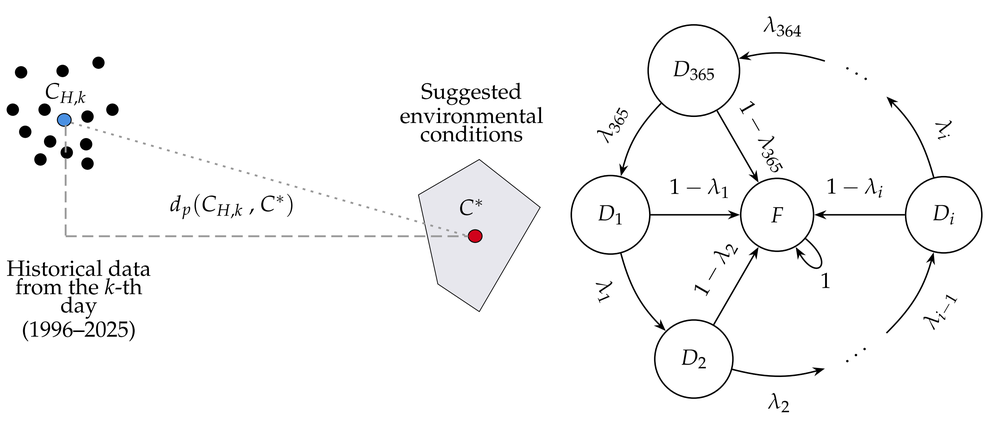
Recalling that given $x = (x_1, \ldots, x_n), y = (y_1, \ldots, y_n) \in \mathbb{R}^n$ and $p \in \mathbb{R}$, $p \geq 1$, the p-distance between x and y is given by

$$d_p(x, y) = \left(\sum_{j=1}^{n} |x_j - y_j|^p \right)^{1/p}. \tag{1}$$

For this work, only $p \in \{1, 2\}$ was considered; i.e., Manhattan and Euclidean distances, respectively, see Figure 2a. Given $p \in \{1, 2\}$, let $d(\cdot, \cdot) := d_p(\cdot, \cdot)$. Therefore, for each $k \in \{1, \ldots, 365\}$

$$\lambda_k = 1 - \frac{d(C_{H,k}, C^*)}{\max\limits_{\ell \in \{1,\ldots,365\}} \{d(C_{H,\ell}, C^*)\}}. \tag{2}$$

denotes the transition probability that will work as weights in the graph shown in Figure 2b.



(a) Goodness-of-sowing of day k, $k \in \{1, \ldots, 365\}$ measured as distance.

(b) Directed graph with weights modeling the sowing process.

Figure 2: From centroid distances to a weighted directed graph for sowing suitability.

In Figure 2b, vertices D_i represent the days of the year (DOY), whereas F represents failure. Let $X = \{X_n\}_{n \in \mathbb{N}_0}$ be a random walk on this directed graph. If $X_0 = D_i$, for some $i \in \{1, \ldots, 365\}$, a viable maize seed is considered to have been sown on the i-th DOY. Otherwise, if $X_0 = F$, the seed is considered non-viable. For $t \geq 1$, if $X_t = D_j$, for some $j \in \{1, \ldots, 365\}$, the germination process is considered successful up to the j-th DOY, whereas $X_t = F$ indicates germination failure due to unfavorable environmental conditions. Furthermore, if $X_t = D_j$, the process continues to the next DOY provided that

$$u_t \leq \lambda_{(j+1) \bmod 365}, \tag{3}$$

where u_t is a realization of $U_t \sim \text{Uniform}(0, 1)$. Otherwise, the process terminates, i.e., $X_{t+1} = F$. Accordingly, 5000 random walks were initialized and simulated for each DOY. In Germany, maize germination takes 16.6 ± 3.1 days [3]. Thus, the probability of emergence following sowing on the i-th DOY is empirically estimated as

$$\widehat{P}_i = \frac{|W(D_i, D_{(i+13) \bmod 365})|}{5000}, \tag{4}$$

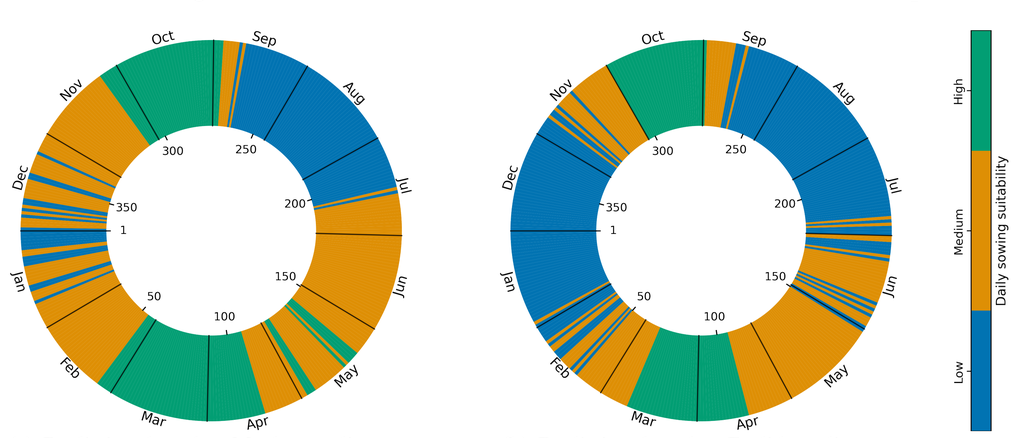
where $|W(D_i, D_{(i+13) \bmod 365})|$ denotes the number of random walks that *at least* reach the minimum germination duration threshold. Under this framework, the i-th DOY is recommended for maize sowing if

$$\widehat{P}_i \geq 0.7^{13}, \tag{5}$$

which corresponds to a minimum environmental suitability threshold of 70% over 13 consecutive days.

Results

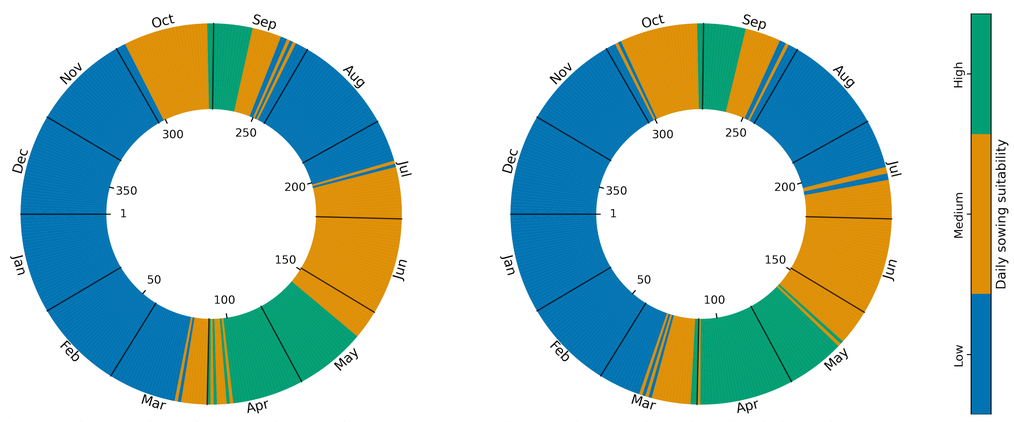
The predicted sowing windows for Brandenburg, Saxony, and Bavaria, obtained using the Manhattan and Euclidean distances, are presented in Figures 3, 4, and 5, respectively. Sowing windows identified according to the criterion in Equation (5) are shown in green. Both distance-based approaches identify periods of environmental suitability for maize sowing, although differences are observed in their temporal extent and location. Remarkably, the Manhattan distance yields narrower sowing windows, whereas the Euclidean distance leads to broader periods.



(a) Predictions based on Manhattan distance.

(b) Predictions based on Euclidean distance.

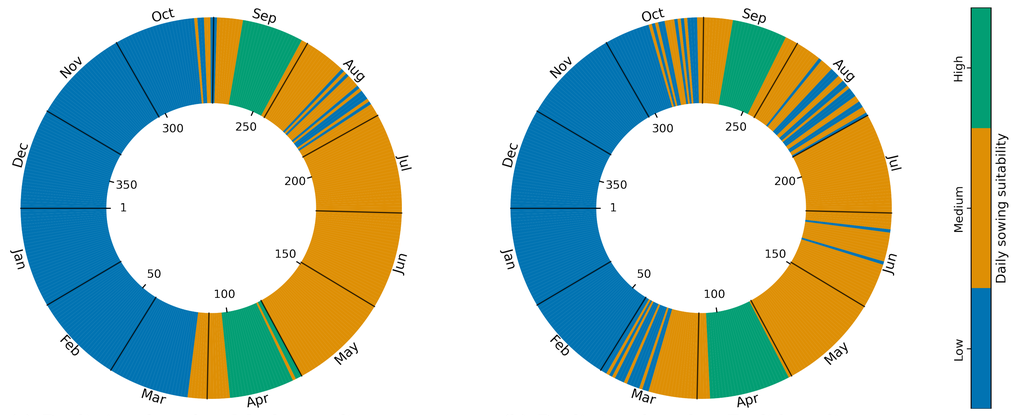
Figure 3: Sowing windows for Brandenburg.



(a) Predictions based on Manhattan distance.

(b) Predictions based on Euclidean distance.

Figure 4: Sowing windows for Saxony.



(a) Predictions based on Manhattan distance.

(b) Predictions based on Euclidean distance.

Figure 5: Sowing windows for Bavaria.

Conclusions

The proposed probabilistic model predicts region-specific sowing windows for Brandenburg, Saxony, and Bavaria, accounting for differences in environmental conditions across regions. The predicted windows during the first half of the year are consistent with the spring-sowing season of maize in Germany. Although additional windows are identified during the second half of the year, these should be interpreted with caution, as subsequent winter conditions may limit crop development. Therefore, their practical suitability requires further assessment beyond the sowing-to-emergence period.

References

[1] M. Amare and B. Balana. Climate change, income sources, crop mix, and input use decisions: Evidence from nigeria. *Ecological Economics*, 211:107892, 2023.

[2] M. Baum, S. Archontoulis, and M. Licht. Planting date, hybrid maturity, and weather effects on maize yield and crop stage. *Agronomy Journal*, 111(1):303–313, 2019.

[3] F. Chmielewski, A. Müller, and E. Bruns. Climate changes and trends in phenology of fruit trees and field crops in germany, 1961–2000. *Agricultural and Forest Meteorology*, 121(1):69–78, 2004.

[4] C. Duden, C. Nacke, and F. Offermann. German yield and area data for 11 crops from 1979 to 2021 at a harmonized spatial resolution of 397 districts. *Scientific Data*, 11(1), Jan. 2024.

[5] DWD Climate Data Center (CDC). Daily grids of mean soil moisture under maize for germany, version v1.0, 2024. Accessed August 5, 2026, at `https://opendata.dwd.de/climate_environment/CDC/grids_germany/daily/soil_moisture/maize/`.

[6] DWD Climate Data Center (CDC). Daily grids of soil temperature at 5 cm depth, 2025. Accessed July 29, 2026, at `https://opendata.dwd.de/climate_environment/CDC/grids_germany/daily/soil_temperature_5cm/`.

[7] DWD Climate Data Center (CDC). Raster data set of mean relative humidity in % for Germany - HYRAS-DE-HURS v6-1, Version v6-1, 2025. Accessed July 31, 2026, at `https://opendata.dwd.de/climate_environment/CDC/grids_germany/daily/hyras_de/humidity/`.

[8] DWD Climate Data Center (CDC). Raster data set of mean temperature in ° C for Germany - HYRAS-DE-TAS v6-1, Version v6-1, 2025. Accessed July 31, 2026, at `https://opendata.dwd.de/climate_environment/CDC/grids_germany/daily/hyras_de/air_temperature_mean/`.

[9] Z. Liu. *Crop Phenology: Timing is everything*, page 257–294. Springer Nature Singapore, 2026.

[10] S. Mason, T. Galusha, and Z. Kmail. Planting date influence on yield of drought-tolerant maize with different maturity classifications. *Agronomy Journal*, 110(1):293–299, 2018.

[11] T. Miedaner and P. Juroszek. Global warming and increasing maize cultivation demand comprehensive efforts in disease and insect resistance breeding in north-western europe. *Plant Pathology*, 70(5):1032–1046, 2021.

[12] R. Patil, M. Lægdsman, J. Olesen, and J. Porter. Growth and yield response of winter wheat to soil warming and rainfall patterns. *The Journal of Agricultural Science*, 148(5):553–566, 2010.

[13] P. Peltonen-Sainio, A. Kangas, Y. Salo, and L. Jauhiainen. Grain number dominates grain weight in temperate cereal yield determination: Evidence based on 30 years of multi-location trials. *Field Crops Research*, 100(2-3):179–188, Feb. 2007.